

MA1521 Calculus for Computing

Lecture Group 2

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Lesson 09: Tuesday 08 September 2026

(Week 5)

Remark. Q08.6 to Q08.8 are leftover exercises from the Friday class in Week 4.

Remark. Q09.1 and Q09.10 are on Sections 13.4, 13.5 of Thomas' Calculus.

Remark. More exercises on Sections 13.4, 13.5, 13.6 and 13.9 will be set for Friday.

Q08.6. Find the maximum and minimum points of the function $f(x, y) = x^2 - 2xy + 2y^2 - 2y$ over the rectangular region $R : -1 \leq x \leq 2, 0 \leq y \leq 2$ in the xy -plane. Hence find the maximum and minimum values of f over the region R .

[Remark: Here you are asked to find the absolute/global maxima and minima of the function f over the region R .

Q08.7. Using the method of least squares, find the line (of the form $y = ax + b$) which best fits the three data points $(-1, 1)$, $(1, 2)$ and $(3, 2)$.

Q08.8. Using the method of least squares, find the quadratic polynomial of the form $y = ax^2 + bx$ which best fits the three data points $(-1, 1)$, $(1, 1)$ and $(2, 0)$.

Q09.1. Consider the function $w = 3x + xy + 2y^2$.

Suppose x is a function of t such that $\frac{dx}{dt} = e^{t^2}$ and

$y = e^t$. Find $\frac{dw}{dt}$, expressing your answer in terms of x

and t . Evaluate also $\left. \frac{dw}{dt} \right|_{t=2}$ if it is given that $x = 3$ when $t = 2$.

Q09.2. Consider the function $w = 2ye^x - \ln z$. Suppose

$$x = \ln(t^2 + 1), \quad y = \tan^{-1}(t), \quad z = e^t.$$

(a) Find $\frac{dw}{dt}$ by using the Chain Rule, and express your answer in terms of t .

(b) Find $\frac{dw}{dt}$ by first expressing w in terms of t .

Q09.3. Suppose the temperature T at any point in the xyz -space is given by

$$T(x, y, z) = e^x - yz + zx,$$

measured in $^{\circ}\text{C}$. Here the coordinates x , y , z are measured in cm. If the position (x, y, z) of an insect at time t , measured in seconds, is given by

$$x = t, \quad y = 2(1 + t^2), \quad z = 3 + t,$$

find the rate of change of temperature experienced by the insect at time $t = 0$. Give your answer in $^{\circ}\text{C}$ per second.

Q09.4. Consider the function $z = 4e^x \ln y$. Suppose

$$x = \ln(u \cos v), \quad y = u \sin v.$$

Use the Chain Rule to find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$, and express your answers in terms of u and v . Evaluate also $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$ at the point $(u, v) = (2, \frac{\pi}{4})$.

Q09.5. A distant planet, Planet ***X***, orbits about a sun, and is orbited by a moon. The amount of rainfall on Planet ***X*** is given by

$$r = f(\theta, h) = h e^{-\frac{1}{\theta^2}},$$

where θ is the average surface temperature on Planet ***X*** and h is the average humidity on the surface of Planet ***X***.

The quantities θ and h are also known to depend on the distance ***s*** of Planet ***X*** from its sun, and the distance ***m*** of Planet ***X*** from its moon. Suppose that when $\theta = 80$ and $h = 10$, you measured that

$$\frac{\partial \theta}{\partial s} = 10, \quad \frac{\partial \theta}{\partial m} = 0.1, \quad \frac{\partial h}{\partial s} = 5, \quad \text{and} \quad \frac{\partial h}{\partial m} = 0.5.$$

Calculate $\frac{\partial r}{\partial s}$ when $\theta = 80$ and $h = 10$.

Q09.6. Suppose $u = f(x, y)$ is a differentiable function on $\mathbb{R}^2$ and all its first-order and second-order partial derivatives are continuous, and $x = r \cos \theta$, $y = r \sin \theta$. Show that

$$\frac{\partial^2 u}{\partial r^2} = f_{xx} \cos^2 \theta + 2f_{xy} \cos \theta \sin \theta + f_{yy} \sin^2 \theta.$$

[Remark: First find $\frac{\partial u}{\partial r}$ in terms of f_x , f_y and θ . You may also use Clairaut's Theorem freely here.]

Q09.7. (a) Find the gradient (i.e. ∇f) of the function $f(x, y) = \sqrt{2x + 3y}$ at the point $(-1, 2)$.

(b) Consider the function

$$f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}} + \ln(xyz).$$

Find ∇f at the point $(-1, 2, -2)$.

Q09.8. Let $f(x, y, z) = xy^2z^3$ and $P = (1, -2, 1)$. Find the rate of change of f at P in the direction $\mathbf{u} = \left\langle \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$.

[Remark: The question is the same as asking to find the directional derivative $D_{\mathbf{u}}f(P)$).

Q09.9. Let a be a positive number. Consider the function

$$f(x, y) = (x^2 + a) e^{2x+3y^2}.$$

It is given that the directional derivative of f at the point $(-6, 2)$ in the direction of the vector $\mathbf{i} + 2\mathbf{j}$ is $216\sqrt{5}$. Find the value of a .

Q09.10. Consider the function $f(x, y) = 2ye^{-x} + 1$ and the point $P(0, -2)$.

(a) Find the directional derivative of f at P in the direction of each of the following vectors:

(i) $\frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j}$, (ii) $12\mathbf{i} + 5\mathbf{j}$.

(b) Find the direction that gives the *largest possible* directional derivative of f at P .

(c) Find the direction along which f decreases most rapidly at P .

(d) Find the directions along which f exhibits no change at P .